Advection Assignment

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1 Introduction

The aim of this project is to find and test advection schemes that perform better than the simplest explicit, linear finite difference schemes and make recommendations of schemes that may be suitable for weather and climate models. Theoretical properties of the alternatives are demonstrated through a set of one-dimensional numerical experiments and are compared with the simple schemes.

2 Basic Schemes

We define schemes for solving the one-dimensional linear advection equation for ϕ:

$$\frac{\partial \phi}{\partial t} = -u\frac{\partial \phi}{\partial x}, \tag{1}$$

on a periodic domain $x \in [0, 1)$ with constant wind speed, u, on a grid of n_x evenly spaced points a distance Δx apart with n_t time steps of length Δt. Grid points are labelled by j and time steps are labelled by n and $\phi_j^{(n)}$ is ϕ at grid point j and time level n. The Courant number is defined as $c = u\Delta t/\Delta x$.

2.1 Forward in Time, Backward in Space (FTBS)

The forward in time, backward in space (FTBS) finite difference scheme is:

$$\phi_j^{(n+1)} = \phi_j^{(n)} - c(\phi_j^{(n)} - \phi_{j-1}^{(n)}). \tag{2}$$

FTBS is first-order accurate in space and time (Weller, 2025). It is stable for $c \in [0, 1]$. Rearrangement of (2) shows that, for $c \in [0, 1]$, $\phi_j^{(n+1)}$ is a convex combination of $\phi_j^{(n)}$ and $\phi_{j-1}^{(n)}$, so FTBS is bounded for $c \in [0, 1]$:

$$\phi_j^{(n+1)} = (1 - c)\phi_j^{(n)} + c\phi_{j-1}^{(n)}. \tag{3}$$

2.2 Forward in Time, Centred in Space (FTCS)

The forward in time, centred in space (FTCS) finite difference scheme is:

$$\phi_j^{(n+1)} = \phi_j^{(n)} - \frac{c}{2}(\phi_{j+1}^{(n)} - \phi_{j-1}^{(n)}). \tag{4}$$

FTCS is second-order accurate in space and first-order in time but is unconditionally unstable (Weller, 2025). Results of FTCS are shown in section 3 but only to help to understand the CTCS results.

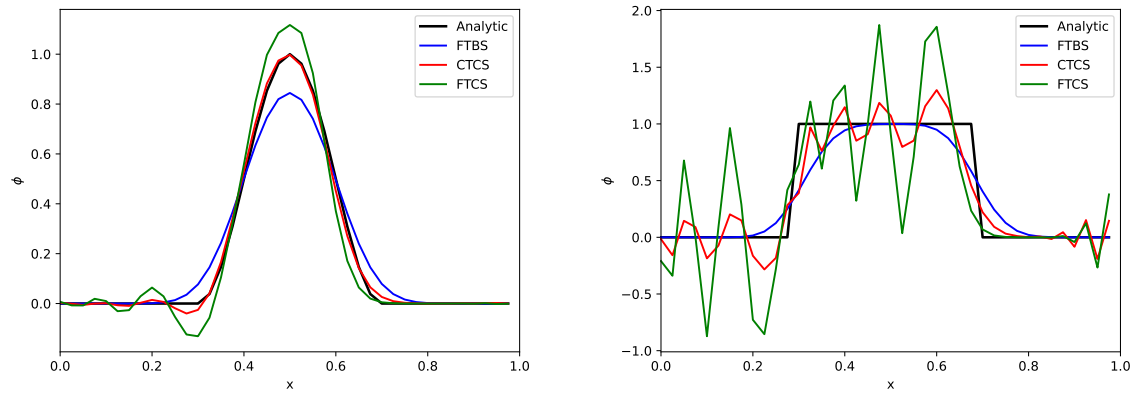


Figure 1: Solution of the advection equation in comparison to the analytic solution for initial conditions (6) and (7) after 10 timesteps with a Courant number of 0.4.

2.3 Centred in Time, Centred in Space (CTCS)

The centred in time, centred in space (CTCS) finite difference scheme is:

$$\phi_j^{(n+1)} = \phi_j^{(n-1)} - c(\phi_{j+1}^{(n)} - \phi_{j-1}^{(n)}). \tag{5}$$

CTCS is second-order accurate in space and time, for the same cost as FTBS, and is stable for $c \in [-1, 1]$ (Weller, 2025). CTCS is a three time-level scheme so requires more storage. There is also a risk that solutions at adjacent times and grid points can display spurious behaviour.

3 Results of the Basic Schemes

Solutions are shown starting from two sets of initial conditions:

$$\phi(x,0) = \begin{cases} \frac{1}{2}\left(1 - \cos 2\pi \frac{x-a}{b-a}\right) & \text{for } a \leq x < b \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

$$\phi(x,0) = \begin{cases} 1 & \text{for } a \leq x < b \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

with $a = 0.1$ and $b = 0.5$. Figure 1 shows results after 10 time steps using a Courant number of 0.4 and 40 points in space in comparison to the analytic solution.

In Figure 1, we observe instability in FTCS in both initial conditions since explicit schemes are unstable for any timestep when applied to hyperbolic problems (LeVeque et al., 2002).

For the continuous cosine initial condition, FTBS remains stable conditional on a Courant number less than or equal to 1 (Courant et al., 1928). However, divergence from the analytic solution was observed due to the numerical diffusion introduced by upwind approximation (Zijlema, 2025). This led to a dampening effect and a loss of total mass when compared to the analytic solution.

Second order accuracy of CTCS yields accurate numerical solution at the given resolution, as well as stability at a Courant number of 0.4 for the continuous initial condition.

For the discontinuous initial condition, we again observe numerical diffusion for the FTBS scheme, and continue to observe stability due to the low enough Courant number. For CTCS, the discontinuities induce oscillations in the numerical solution due to the dispersivity of the scheme.

4 Alternative Scheme

The spectral method for the 1D advection equation is defined as follows (Weller, 2026). By truncating ϕ in its Fourier basis, we have the approximation for even N

$$\phi_N(x,t) := \sum_{k=-N/2}^{N/2-1} \widehat{\phi}_k(t) \exp(2\pi i k x).$$

Substituting ϕ_N for ϕ in equation (1) yields

$$\sum_{k=-N/2}^{N/2-1} \frac{d\widehat{\phi}_k}{dt} \exp(2\pi i k x) = -2\pi i u \sum_{k=-N/2}^{N/2-1} k\widehat{\phi}_k \exp(2\pi i k x) \tag{8}$$

so that in order for $\phi_N(\cdot,t)$ to be a Galerkin approximation to $\phi(\cdot,t)$ in span$\{\exp(2\pi i k x)\}_k$ at each time t, by orthogonality of Fourier modes it is enough that

$$\frac{d\widehat{\phi}_k}{dt} = -2\pi i k u \widehat{\phi}_k, \quad -\frac{N}{2} \le k \le \frac{N}{2} - 1. \tag{9}$$

The update over each timestep can be solved exactly as

$$\widehat{\phi}_k(t+\Delta t) = \widehat{\phi}_k(t) \exp(-2\pi i k u \Delta t). \tag{10}$$

An inverse Fourier transform applied to (10) yields our Spectral approximation to ϕ at the new timestep.

The main advantage we expect to observe is that convergence of ϕ_N to ϕ wil be exponential in N for smooth analytic initial conditions (Hussaini and Zang, 1987), allowing us to cheaply construct accurate approximations to the solution using a relatively large stepsize. Moreover, since we were able to solve for the timestep update exactly we expect to observe stability unconditional on the timestep allowing for larger timesteps compared to the conditionally stable CTCS and FTBS schemes.

5 Results

5.1 Snapshots of the Solutions

The results of the new scheme for the setup defined in section 3 are shown in figure 2 which are compared with CTCS and FTBS.

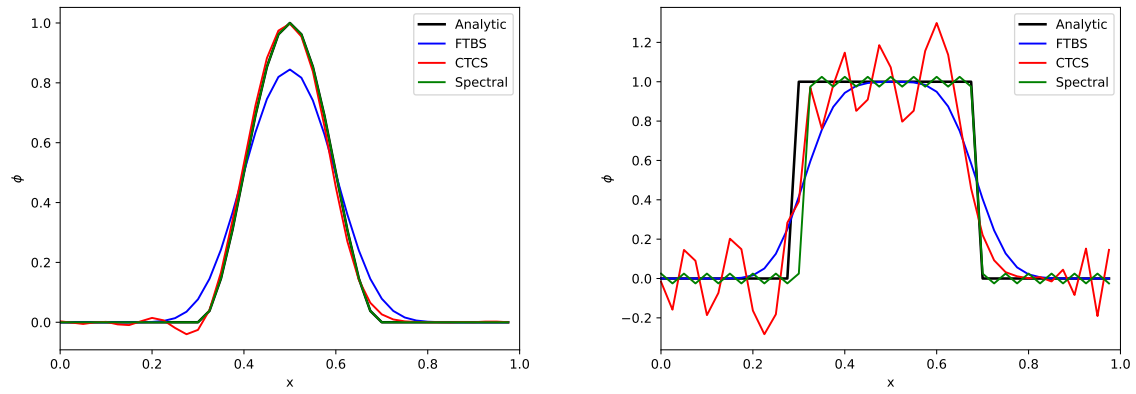


Figure 2: Solution of the advection equation in comparison to the analytic solution for initial conditions (6) and (7) for the scheme defined in section 4 after 10 timesteps.

In figure 2, we observe numerical solution from the spectral method being very close to the analytic solution for the cosine initial condition. This is likely due to the exact time update (10) which preserves the $l^2[0,1)$ norm of the numerical solution over time, as well as small spatial discretisation error at $N = 40$. The numerical solution therefore exhibits the same mass preserving property as the analytic solution, allowing the spectral approximation to remain close to the analytic solution over time without numerical diffusion or instability.

For the discontinuous initial condition, some oscillations are introduced in the spectral solution due to the Gibbs phenomenon (Boyd, 2001). Although errors still appear smaller than for the other finite difference methods and the scheme appears stable in this case, we cannot in general expect better than sublinear in N l^2 convergence for the spectral method applied to discontinuous initial conditions without taking postprocessing steps (Piotrowska et al., 2019). Moreover, we cannot expect l^∞ convergence as the peak amplitude of Gibbs oscillations does not decrease with N.

5.2 Verification of Theoretical Property: Exponential Convergence

We claim that convergence in $N := \frac{1}{\Delta x}$ is exponential for the spectral method with smooth analytic initial conditions, while remaining algebraic for the finite difference schemes. That is,

$$||\phi_N - \phi||_{l^2}^2 \leq C_1 \exp(-C_2 N)$$

for some $C_1, C_2 > 0$ for ϕ_N defined via the spectral method. In order to verify this theoretical property, we require a smooth initial condition unlike those used so far. For this, we choose

$$\phi(0,x) = \exp(\sin(2\pi x)), \quad 0 \leq x < 1 \tag{11}$$

which remains periodic over the domain and has infinitely many nonzero Fourier coefficients. By varying Δx, we vary the number of Fourier modes N used for our discrete Fourier transform. We choose $N = 3, 8, 13, ..., 33$, and keep our domain as $[0,1]$.

In figure 3, we observe that the logarithmic l^2 error of the spectral solution appears to decay linearly with the number of Fourier modes used, supporting the theoretical property of

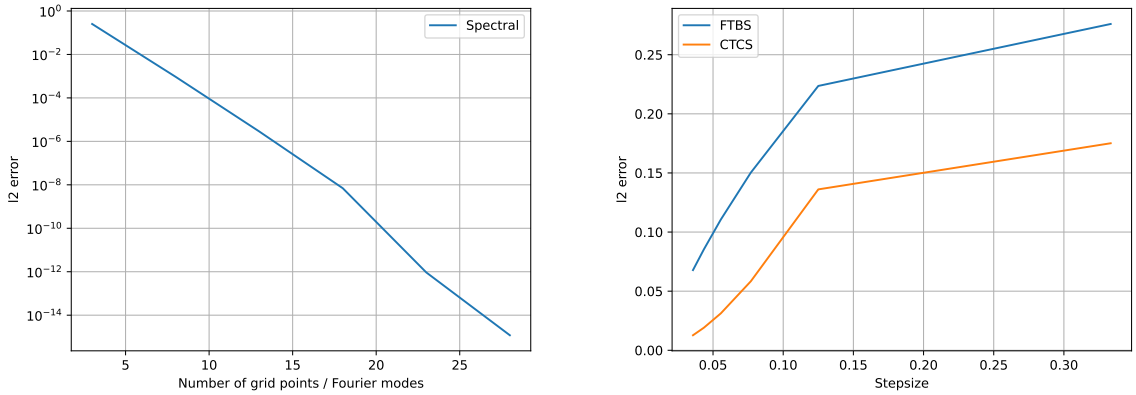


Figure 3: l^2 error scaling after 20 timesteps for the Spectral and finite difference advection schemes with initial condition (11)

exponential convergence. Moreover, we observe error scaling with Δx for the finite difference schemes to be algebraic as claimed - confirming improved error scaling for the Spectral advection scheme.

5.3 Verification of Theoretical Property: l^2 Norm Conservation

We claim that the spectral method will conserve the l^2 norm of the discrete solution over time, mirroring the behaviour of the true solution. In comparison, we expect the difference in l^2 norm of numerical solutions obtained using the conditionally stable CTCS scheme to vary but remain bounded. For the diffusive FTBS scheme, we expect the norm of the numerical solution to decrease with time, while for the unstable FTCS scheme, we expect the difference in l^2 norm between the analytic and numerical solution to grow with time. This is due to differentiation and timestepping used in the spectral method being exact in the Fourier domain, meaning that errors arise only from truncation of the Fourier basis.

For this experiment, the smooth analytic initial condition (11) was used to ensure no pathological behaviour arose from nonsmooth initial conditions. Moreover, we used the parameters $N = 40$, with 20 timesteps over a time period of 0.2 with u$= 1$. This gave a Courant number of 0.4 ensuring that conditionally stable schemes were stable in our regime.

After running our experiment, we observed exact norm conservation of the spectral method as expected, as well as the anticipated bounded variation in the CTCS norm change and growth in the unstable FTCS solution norm. We also observed norm decay from the FTBS scheme, due to numerical diffusion.

6 Summary and Conclusions

Our experiments demonstrate the behaviour of the basic finite difference schemes and the spectral method. FTBS is stable and easy to implement, but it is only first-order accurate and introduces substantial numerical diffusion. FTCS is not suitable for pure advection because it is unconditionally unstable. CTCS is second-order accurate and less diffusive than FTBS, but it

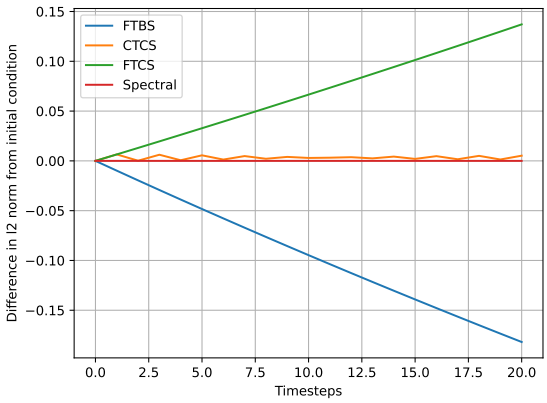


Figure 4: Change in l^2 norm relative to the true solution as a function of time for FTBS, FTCS, CTCS, and spectral methods.

can generate dispersive oscillations near discontinuities and requires three time levels so implementation is more complex.

The Fourier spectral method is highly effective for smooth periodic solutions. It represents the spatial derivative accurately within the truncated Fourier space and preserves the amplitude of every resolved Fourier mode with an exact time update. Our tests confirmed exponential convergence in the number of spatial discretisation steps - equivalent to the number of Fourier modes used. Our stability experiment also confirmed conservation of the l^2 norm, mirroring the behaviour of the true solution.

The limitations of the spectral method arose when considering discontinuous initial conditions. The square wave produced Gibbs oscillations, and the exponential convergence property does not hold in general for non-analytic initial conditions. Therefore, the spectral method appears to perform better than finite difference methods for smooth, periodic models of weather and climate, especially over longer time periods. Where discontiuities occur, postprocessing may be required for effective use of the spectral method, or other schemes such as finite volumes may need to be considered.

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